

The Projection–Admissibility Principle

Descent, Recovery, and Structural Constraints on Effective Description

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Abstract

We replace the former noninvertibility obstruction by a typed theory of projection, descent, recovery, and admissibility. For an upper evolution Φ_t , an initial reduction P_0 , and a final reduction P_t , four different questions arise. A right section selects one compatible upper representative; a left decoder recovers the actual upper input; an autonomous reduced map exists exactly when upper evolution respects the initial equivalence classes; and an effective merger is noninjectivity of that reduced map. Noninjectivity of a cross-level projection does not forbid a right section. We prove the correct factor-through theorem, a valid no-right-section criterion for a genuinely contractive reduced self-map, a section-conditioning diagnostic, and locality descent for fiberwise coherent compression of a local operator net. Probability requires an upper measure and disintegration; entropy, irreversibility, and an arrow of time require additional structures. Applications to open systems, Wilsonian reduction, exterior gravity, and MTT are consequently reconstructions or conditional realizations rather than corollaries of noninjectivity alone.

Revision note for this edition

Supersedes. The original *Projection–Admissibility Principle: Structural Constraints on Effective Physical Description*.

Reason. Noninjectivity of a cross-level projection was incorrectly used to rule out a right section and to infer several unrelated physical effects.

Resolution. Version 2 separates right sections, left decoders, autonomous descent, and effective mergers, and replaces the obstruction claim by the correct factor-through and conditioning theorems.

Retained result. Projection and admissibility remain useful structural organizers once every map is typed.

Remaining boundary. Probability, entropy, irreversibility, and physical recovery require measures, dynamics, and observable-specific hypotheses.

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1 How to Read the Principle

A projection deliberately forgets distinctions. The central issue is not whether information was forgotten, but which later questions can still be answered from what remains. Four questions recur throughout the paper:

Choose a representative.

Required equation: $T_t S_t = \text{id}$. Failure means that some final reduced states have no admissible upper representative, or none with the requested regularity.

Recover the actual input.

Required equation: $D_t T_t = \text{id}$. Failure means that distinct upper inputs have become indistinguishable at the final reduced level.

Evolve autonomously.

Required equation: $F_t P_0 = P_t \Phi_t$. Failure means that the initial reduced state omits distinctions that influence the later reduced state.

Detect a merger.

Required relation: $F_t(y) = F_t(y')$ for $y \neq y'$. Its occurrence means that distinct initial reduced states have the same final reduced image.

The equations point in different directions and therefore cannot be replaced by one slogan about invertibility.

1.1 Object picture: what the reduced state remembers

For the projection $P_0(x, h) = x$, the coordinate h is hidden. If upper evolution changes the visible coordinate by an amount depending on h , two upper states represented by the same x can have different visible futures. Autonomous descent then fails because the reduced state has forgotten something needed for prediction. A representative section can still choose, for example, $h = 0$; that choice does not recover the hidden value that was actually present.

This memory picture is useful across the applications. Initial system–environment correlations, eliminated ultraviolet couplings, interior boundary data, and complementary coherent modes can all act as an h -type variable. Whether they matter is decided by the factor-through condition, not by the mere fact that a projection was used.

1.2 Argument map

Sections 2–4 type the maps and prove the descent/recovery classification. Section 5 gives the genuine diameter obstruction. Sections 6 and 7 treat stable continuation and chart boundaries. Sections 8 and 9 explain which measure and temporal data must be added for stochastic or irreversible interpretations. Section 10 proves the separate locality-descent statement. The final applications then show how the same distinctions change the interpretation of open systems, Wilsonian reduction, exterior gravity, and MTT.

2 Scope and correction of the former obstruction

The previous version argued that a noninjective map cannot have a right inverse. That statement is false. A surjection may be highly noninjective and still admit a section. For example,

$$r : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad r(x, z) = x, \quad s(x) = (x, 0)$$

satisfies $r \circ s = \text{id}_{\mathbb{R}}$.

The mistake was a type error. A right inverse chooses one preimage; it does not recover the preimage that was actually supplied. Actual microscopic recovery is a left-inverse condition. Autonomous effective evolution is a third condition, namely factorization through the initial reduction. This paper keeps those objects separate.

The results are structural set-theoretic, topological, metric, measure-theoretic, or operator-algebraic theorems according to the hypotheses stated in each section. No probability, entropy, Hilbert structure, geometry, or physical time is inferred from a bare map of sets.

3 Typed projection and evolution data

Let A be an admissible upper state domain and let

$$\Phi_t : A \rightarrow X_t$$

be an upper evolution map. Invertibility of Φ_t is not needed for the basic typing results. Let

$$P_0 : A \rightarrow Y_0, \quad P_t : \Phi_t(A) \rightarrow Y_t$$

be surjective reduction maps onto their declared effective images, and define the cross-level output map

$$T_t = P_t \circ \Phi_t : A \rightarrow Y_t.$$

The fibers of P_0 define the initial effective equivalence relation

$$x \sim_0 x' \iff P_0(x) = P_0(x').$$

Definition 3.1 (Representative section). A representative section for T_t is a map $S_t : Y_t \rightarrow A$ satisfying

$$T_t \circ S_t = \text{id}_{Y_t}.$$

It chooses one upper state compatible with each final effective state.

Definition 3.2 (Exact upper decoder). An exact decoder is a map $D_t : T_t(A) \rightarrow A$ satisfying

$$D_t \circ T_t = \text{id}_A.$$

It recovers the actual upper input from the final effective output.

Definition 3.3 (Autonomous reduced evolution). An autonomous reduced evolution is a map $F_t : Y_0 \rightarrow Y_t$ satisfying

$$F_t \circ P_0 = P_t \circ \Phi_t.$$

Definition 3.4 (Effective merger). If F_t exists, an effective merger occurs when distinct initial effective states $y \neq y'$ satisfy $F_t(y) = F_t(y')$.

These four definitions have different domains and equations. They are not interchangeable notions of inversion.

4 Projection–descent and recovery theorem

Theorem 4.1 (Projection–descent and recovery). *For the typed data above:*

1. *A set-theoretic representative section S_t exists if T_t is surjective and the relevant choice principle is available. A continuous, measurable, local, or Lipschitz section requires a theorem in that category.*
2. *An exact decoder $D_t : T_t(A) \rightarrow A$ exists only if T_t is injective. Conversely, if T_t is injective, its inverse on $T_t(A)$ is an exact decoder.*
3. *An autonomous reduced evolution F_t exists if and only if*

$$P_0(x) = P_0(x') \implies P_t(\Phi_t x) = P_t(\Phi_t x') \tag{4.1}$$

for all $x, x' \in A$. When it exists, F_t is unique.

4. *If (4.1) holds and there are $x, x' \in A$ with $P_0(x) \neq P_0(x')$ but $P_t(\Phi_t x) = P_t(\Phi_t x')$, then F_t is noninjective. The prior effective state cannot be decoded uniquely from the final effective state.*

Proof. The first item is the definition of a section of a surjection. If $D_t T_t = \text{id}_A$ and $T_t(x) = T_t(x')$, applying D_t gives $x = x'$, proving the second item. For the third, necessity follows by applying F_t to equal initial effective states. Under (4.1), define

$$F_t(P_0 x) := P_t(\Phi_t x).$$

The implication makes the value independent of the representative; surjectivity of P_0 gives existence and uniqueness. The last item follows directly from the definition of F_t . $\square$

Corollary 4.2 (Noninjectivity does not obstruct representative selection). *Noninjectivity of T_t rules out an exact decoder, not a representative section. Surjectivity is the set-theoretic gate for a right section.*

The load-bearing part of Theorem 4.1 is item 3. It asks whether the upper evolution maps every initial projection fiber into one final projection fiber. If so, all upper representatives of one reduced state give the same reduced future and F_t is well defined. If not, choosing a section can manufacture one representative-dependent trajectory, but it cannot turn that trajectory into an autonomous law on Y_0 .

5 A valid no-right-section obstruction

There is a correct obstruction when the map in question is a self-map of the same reduced space and its image is provably too small.

Theorem 5.1 (Finite-diameter contraction obstruction). *Let (Y_A, d) have finite positive diameter D , and let $G : Y_A \rightarrow Y_A$ satisfy*

$$d(Gy, Gy') \leq \kappa d(y, y') + c\varepsilon, \quad 0 \leq \kappa < 1, \quad c\varepsilon \geq 0.$$

Then

$$\text{diam } G(Y_A) \leq \kappa D + c\varepsilon.$$

If $(1 - \kappa)D > c\varepsilon$, then G is not surjective and therefore has no right section $S : Y_A \rightarrow Y_A$ with $G \circ S = \text{id}_{Y_A}$.

Proof. Take the supremum of the displayed inequality over all pairs. Under the strict condition the image diameter is smaller than D , whereas a surjective image would equal Y_A and have diameter D . $\square$

This theorem concerns a reduced self-map $G : Y_A \rightarrow Y_A$. It does not apply to a cross-level map $T_t : A \rightarrow Y_t$ merely because that map is noninjective. Moreover, the additive-error inequality is not a Banach contraction. Banach requires a genuine Lipschitz constant below one on a complete invariant domain.

6 Stable representative continuation

Suppose A, Y_t are metric spaces. For all Lipschitz right sections define the best section condition number

$$\kappa_{\text{sec}}(t) := \inf\{\text{Lip}(S_t) : T_t S_t = \text{id}_{Y_t}\},$$

with $\kappa_{\text{sec}}(t) = +\infty$ if no Lipschitz section exists.

Proposition 6.1 (Section-conditioning diagnostic). *If $\kappa_{\text{sec}}(t)$ remains bounded on an interval and a chosen family of sections attains a uniform bound, representative selection is uniformly Lipschitz on that interval. Blow-up of κ_{sec} or loss of surjectivity is an obstruction to such stable continuation. It is not, by itself, a physical singularity, entropy law, or stochastic transition.*

For spectral projections, a parallel diagnostic is failure of bounded norm-resolvent continuation of the selected Riesz projector. These two diagnostics are related only when a realization proves the relation.

7 Admissibility without automatic selection

Let $m_j(t, x)$ be declared continuous margins and define

$$A_t^\delta = \{x : m_j(t, x) \geq \delta \text{ for every } j\}.$$

The margins may record operator domains, spectral separation, complementary damping, leakage, contraction, truncation error, section conditioning, or physical hyperbolicity. A first exit from A_t^δ means that at least one declared description condition fails.

Exit does not imply that T_t has no set-theoretic section; the failed margin must be identified. Nor does exit select a new state. One must stop the reduced description, derive continuation from upper dynamics, or add a hybrid reset law. A reset is new continuation data and must prove conservation, measurability, and any assigned probabilities.

8 Measure-dependent stochastic reduction

Projection does not create a probability measure. A stochastic reduced law can be induced only after preparation or invariant measure data are supplied.

Theorem 8.1 (Measure-dependent reduced kernel). *Let A, Y_0, Y_t be standard Borel spaces, let P_0 be measurable, and let μ be a probability measure on A . Let $\{\mu_y\}$ be a regular conditional distribution of x given $P_0(x) = y$. Then*

$$K_t(y, B) := \mu_y(\{x : P_t(\Phi_t x) \in B\})$$

defines a Markov kernel from Y_0 to Y_t , up to the usual $(P_0)_\# \mu$ -null sets. If autonomous descent holds, then

$$K_t(y, \cdot) = \delta_{F_t(y)}$$

for almost every y .

Proof. Regular conditional distributions exist on standard Borel spaces. Measurability and countable additivity pass through the measurable preimage defining K_t . Under descent, $P_t \Phi_t x = F_t(P_0 x)$ is constant on each conditional fiber. $\square$

Different upper measures, even with the same projection fibers, may induce different kernels. Fiber cardinality alone does not determine weights and does not imply the Born rule.

9 Information, entropy, and temporal direction

Noninjectivity states only that some distinctions are absent from the reduced state. Quantitative information requires a sigma-algebra, coding, metric, or probability measure. Entropy requires a specified entropy functional and state. Monotone entropy production requires a dynamical theorem such as data processing for a declared stochastic channel, a coarse-graining inequality, or a thermodynamic balance law.

Likewise, failure of an exact decoder does not by itself establish an arrow of time. A temporal arrow requires an oriented evolution family and an asymmetric property such as a semigroup without inverse in the chosen category, a monotone Lyapunov/entropy functional, or a boundary condition. A representative section is not a reverse-time evolution.

Universality also needs more than equivalence classes. It requires stability of reduced laws under a declared class of microscopic perturbations, for example the basin-local contraction estimate of the corrected Foundation and Fixed Points spine.

10 Locality descent under coherent compression

Let $\pi : M_{10} \rightarrow Y_4$ be the canonical physical bundle and let

$$P = \int_{Y_4}^{\oplus} P_x d\mu(x)$$

be a decomposable fiberwise coherent projector. Suppose $O \mapsto \mathcal{A}_{10}(\pi^{-1}O)$ is an upper local net. Define

$$\mathcal{A}_{10}^P(O) = \{A \in \mathcal{A}_{10}(\pi^{-1}O) : [A, P] = 0\}$$

and

$$\mathcal{A}_4(O) = \{PAP|_{\text{Ran } P} : A \in \mathcal{A}_{10}^P(O)\}.$$

Theorem 10.1 (Fixed-point locality descent). *If the upper net is isotonic, then $\mathcal{A}_4$ is isotonic. If upper observables assigned to spacelike separated base regions commute, then their compressed coherent observables commute. Thus microcausality descends on the P -compatible subalgebra.*

Proof. Upper inclusion immediately gives compressed inclusion. For compatible A, B , commutation with P gives

$$[PAP, PBP]|_{\text{Ran } P} = P[A, B]P|_{\text{Ran } P},$$

which vanishes for spacelike separated upper observables. $\square$

This theorem does not imply state factorization. Entangled or otherwise nonfactorizing states may restrict to the descended net. A projection nonlocal over the base, or an observable not preserving $\text{Ran } P$, lies outside this theorem.

Thus coherent compression preserves locality only for the algebra on which the displayed commutation calculation is legal. The theorem transports an existing upper microcausal relation; it does not create spacetime locality from an arbitrary projector and does not force the compressed state to be unentangled.

11 Corrected realizations

11.1 Open quantum systems

For a system and environment, partial trace

$$P(\rho_{SE}) = \text{Tr}_E \rho_{SE}$$

is noninjective but surjective onto system density matrices when an environment state is available. The assignment $S(\rho_S) = \rho_S \otimes \sigma_E$ is a representative section. It is not an exact decoder of the actual correlated state ρ_{SE} .

A state-independent reduced channel exists only under assignment and compatibility conditions. Initial system–environment correlations can make the future reduced state depend on more than ρ_S , violating the descent criterion. Probabilities are already supplied by the quantum state and Born rule; they are not derived from noninjectivity of partial trace.

11.2 Wilsonian reduction

Integrating out high-frequency modes maps UV actions or measures to lower-scale effective data. Distinct UV inputs may have the same IR image, so exact UV decoding generally fails. A representative UV completion, when it exists, is a section and is nonunique. Autonomous RG evolution requires a selected theory space and closure or controlled truncation; it does not follow from coarse graining alone. Universality requires a fixed-point/basin stability theorem.

11.3 Exterior gravity

Restriction of a global solution to an exterior domain can be noninjective. An exterior-to-global extension, where constraint-compatible extensions exist, is a representative section rather than recovery of the actual interior. Autonomous exterior evolution requires boundary conditions, flux data, and a well-posed domain. Horizon area or entropy is not obtained from noninjectivity alone.

11.4 Modal Triplet Theory

In the corrected MTT Foundation, P is the joint coherent spectral projector and R_τ is the stabilization flow. An autonomous coherent map exists exactly when PR_τ is constant on the relevant initial P -fibers. Exact invariance is a sufficient special case. FP I–VI provide conditional fixed-point, damping, curved-projector, covariance, and admissibility results.

The internal gap and Riesz-projector continuation can support stable coherent reduction. They do not by themselves derive a measure, Born weights, entropy, a reset outcome, quantum field theory, gravity, or cosmology. MTT is therefore a conditional realization of the typed framework only to the extent that each gate is verified.

12 Scoped projection–admissibility theorem

Theorem 12.1 (Scoped principle). *For typed upper evolution and reductions:*

1. *representative selection, exact recovery, autonomous descent, and effective merger obey the separate conditions in Theorem 4.1;*
2. *a reduced self-map has no right section when its image diameter is strictly smaller than its state-space diameter;*
3. *stable representative continuation can be tested by section conditioning or, in spectral realizations, bounded projector continuation;*
4. *an upper measure and regular conditional distributions induce a reduced Markov kernel; and*
5. *upper locality descends to the coherent P -compatible compressed net.*

No probability rule, entropy production, arrow of time, universality class, geometry, or Hilbert structure follows from noninjectivity alone.

13 Conclusion

Projection and admissibility remain useful organizing ideas once their maps are typed correctly. The corrected principle says precisely when effective dynamics descends, what a section can and cannot recover, when a reduced map truly loses a right section, and which additional

measure, stability, and locality data support stronger conclusions. This narrower theorem is more useful than the former universal obstruction because each downstream claim now has a checkable mathematical gate.

References

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Computational Evidence and Reproducibility

The numerical and machine-verifiable claims used by this paper are archived in the curated repository, <https://github.com/PeterNero/mtt-results-repro>. The mapped authority/result identifiers are `no result rows mapped`. Claim tiers in that capsule distinguish exact derivation, certified numerics, profile replay, conditional results, and open obligations.